

Analysis of Operational and Clinical Patterns in MIMIC-III



Insights into Workforce, Efficiency, and Cost Allocation



Workforce



Efficiency

Cost Allocation



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Dataset Overview



Goal: Analyze Electronic Health Record (EHR) data to uncover patterns in hospital operations and patient intake.



Data Source: MIMIC-III Clinical Database (Demo Version)

Unique Tables Selected:

CAREGIVERS



Workforce roles

CALLOUT



ICU discharge efficiency

CPTEVENTS



Financial cost centers

ADMISSIONS



Patient flow and diagnosis

Tools: Python (Pandas, Seaborn, Matplotlib)



pandas



seaborn



Visual 1: Workforce Composition

- Caregiver Role Distribution

- Key Insights:



The ICU workforce is heavily dominated by Registered Nurses (RN) and Residents



This highlights the labor-intensive nature of critical care, which requires monitoring by nursing staff

- How I Built It (Replication Steps):



Source: Loaded CAREGIVERS.csv



Filter: I identified the top 10 most common job titles using `.value_counts().head(10)`



Clean: I used a list comprehension `[x if x in top_jobs else 'Other'...]` to group rare job titles into an 'Other' category to declutter the chart

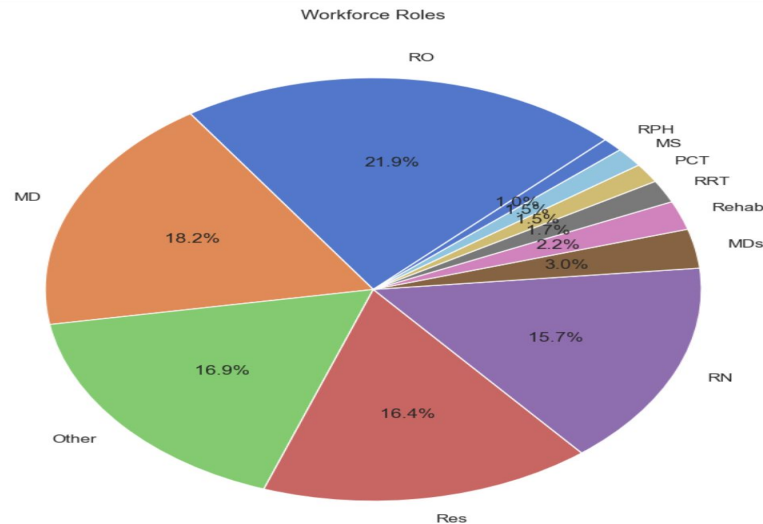


Plot: Created a Pie Chart using `plt.pie()` for distinct segments

```
# 2) visual 1: job titles (pie chart)
# top 10 and placing rest in others category
top_jobs = df_cg['label'].value_counts().head(10).index.tolist()
df_cg['job_group'] = [x if x in top_jobs else 'Other' for x in df_cg['label']]

# totals for chart
job_stats = df_cg['job_group'].value_counts()

# pie chart
plt.figure(figsize=(10, 10))
plt.pie(job_stats, labels=job_stats.index, autopct='%1.1f%%', startangle=45,
        colors=sns.color_palette('muted'))
plt.title('Workforce Roles')
plt.show()
```



Visual 2: Operational Efficiency

- **ICU Discharge Response Time** 🕒

- **Key Insights:**

- This metric tracks the delay between a discharge request being created and acknowledged
- The distribution is skewed heavily to the left, proving the ICU staff is highly efficient

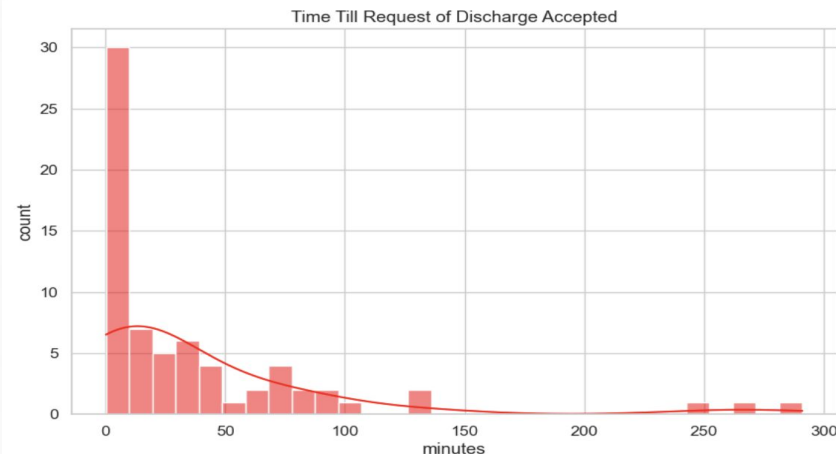
- **How I Built It (Replication Steps):** 🛠️

- **Source:** Loaded CALLOUT.csv
- **Calculate:** Created a delay column by subtracting createtime from acknowledgegetime, converting to minutes: `.dt.total_seconds() / 60`
- **Refine:** Filtered for positive values under 5 hrs to remove outliers
- **Plot:** Used `sns.histplot()` with `kde=True` to overlay a density curve, showing the probability distribution

```
# 3) visual 2: response delay (histogram)
# calculate delay time
df_call['delay'] = (df_call['acknowledgegetime'] - df_call['createtime']).dt.total_seconds() / 60

# filtering out outliers (just looking at under 5 hours)
# we don't want negative numbers or extreme delays
df_clean = df_call[(df_call['delay'] > 0) & (df_call['delay'] < 300)]

# plotting
plt.figure(figsize=(10, 6))
sns.histplot(data=df_clean, x='delay', bins=30, color='red', kde=True)
plt.title('Time Till Request of Discharge Accepted')
plt.xlabel('minutes')
plt.ylabel('count')
plt.show()
```





Visual 3: Financial Resource Allocation

Procedures by Cost Center

Procedures by Cost Center

Key Insights:

- 💡 The **ICU Cost Center** generates significantly more billed procedures than the Respiratory center
- 💡 This suggests that general ICU management accounts for the **bulk of financial volume**

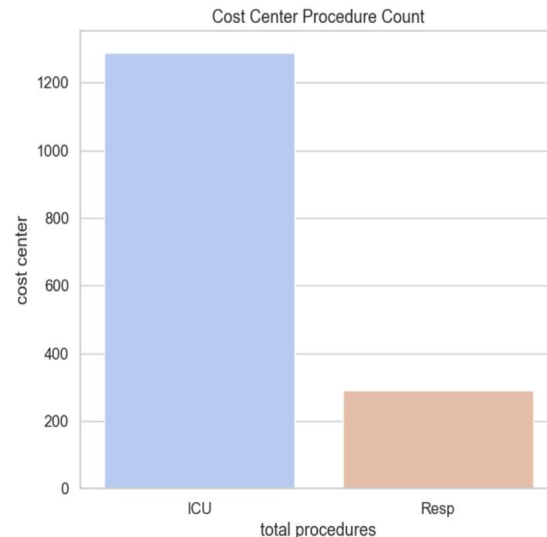
How I Built It (Replication Steps):

- 🗄️ **Source:** Loaded CPTEVENTS.csv
- ⚙️ **Aggregate:** Used `df['costcenter'].value_counts()` to count procedures per department
- 📊 **Plot:** Mapped the data to a vertical bar chart using `sns.barplot()`
- 🎨 **Style:** Used the coolwarm palette and `dodge=False` to distinctively color the bars by department

```
# 4) visual 3: cost centers (bar chart - vertical)
# counting the procedures
dept_activity = df_cpt['costcenter'].value_counts()

# plot
plt.figure(figsize=(7, 6))
sns.barplot(x=dept_activity.index, y=dept_activity.values, hue=dept_activity.index, palette='coolwarm', dodge=False)

plt.title('Cost Center Procedure Count')
plt.ylabel('cost center')
plt.xlabel('total procedures')
plt.legend([], [], frameon=False)
plt.show()
```





Visual 4: Clinical Pathology

Prevalence of High-Risk Diagnoses

Prevalence of High-Risk Diagnoses

Key Insights:



Performed text analysis on the admission records to find high-risk keywords



Sepsis and **Heart** issues are the primary drivers of admission, aligning with critical care pathology

How I Built It (Replication Steps):



Source: Loaded ADMISSIONS.csv



Search: Defined a list of keywords (SEPSIS, PNEUMONIA) and used a dictionary comprehension with `.str.contains().sum()` to count occurrences



Structure: Converted the dictionary to a Pandas DataFrame and sorted by count



Plot: Used `sns.barplot()` with `orient='h'` (Horizontal) to make the text easy to read

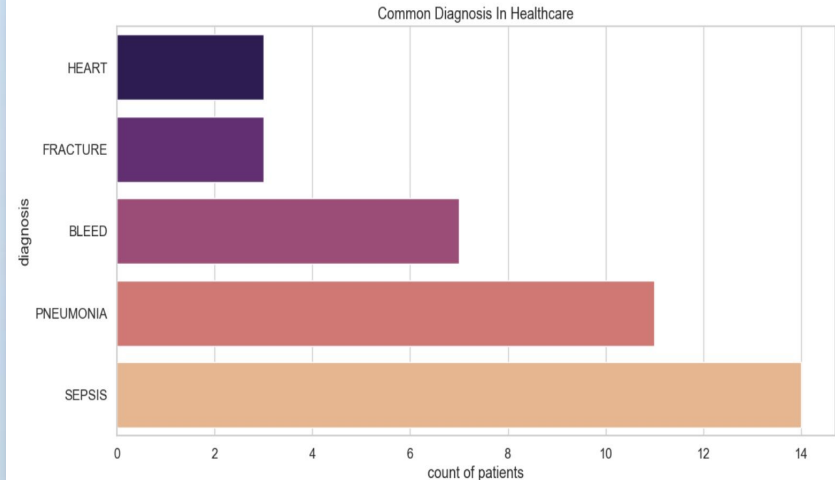
```
# 5) visual 4: common diagnosis (bar chart - horizontal)
# keywords in data
search_terms = ['SEPSIS', 'HEART', 'PNEUMONIA', 'BLEED', 'FRACTURE']

# get the counts
results = {term: df_adm['diagnosis'].str.contains(term, na=False).sum() for term in search_terms}

# sorting dataframe
final_df = pd.DataFrame(list(results.items()), columns=['disease', 'total']).sort_values('total')

# plot
plt.figure(figsize=(13, 6))
sns.barplot(data=final_df, x='total', y='disease', hue='disease', palette='magma', dodge=False)

plt.title('Common Diagnosis In Healthcare')
plt.xlabel('count of patients')
plt.ylabel('diagnosis')
plt.legend([], [], frameon=False)
plt.show()
```





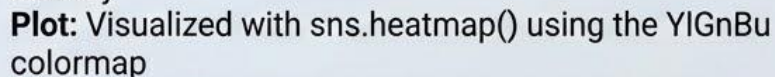
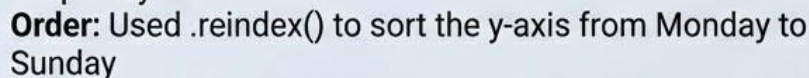
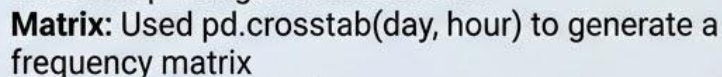
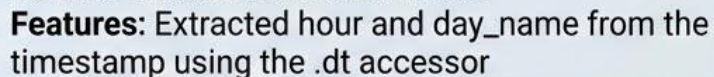
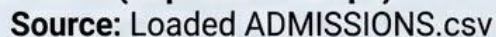
Hospital Intake Heatmap (Bonus)

- ### Key Insights:



- Reveals **"hot spots"** for admissions (**weekday, evenings**) that simple daily counts miss, helping in **staff scheduling**

How I Built It (Replication Steps):

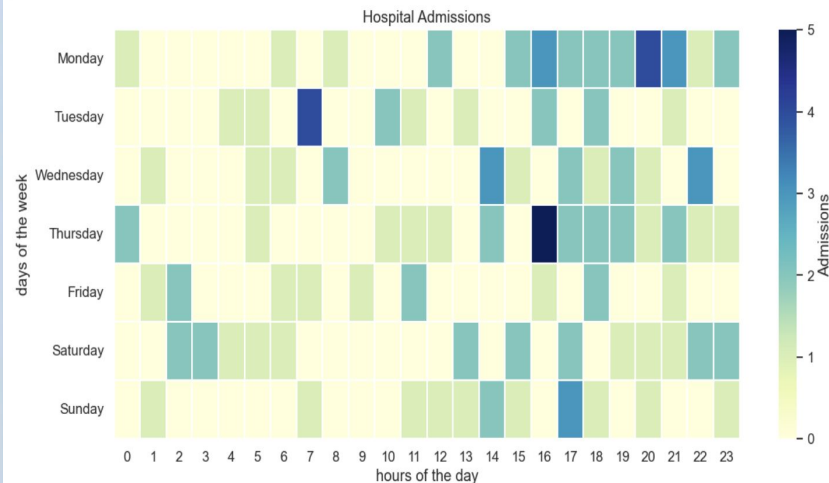


```
# 6) visual 5: busy times heatmap (novel data visual for bonus credit)
# getting hour/day name from timestamp
df_adm['adm_hour'] = df_adm['admittime'].dt.hour
df_adm['adm_day'] = df_adm['admittime'].dt.day_name()

# create frequency grid
adm_grid = pd.crosstab(df_adm['adm_day'], df_adm['adm_hour'])

# ordering days of week
week_order = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday']
adm_grid = adm_grid.reindex(week_order)

# plot
plt.figure(figsize=(13, 6))
sns.heatmap(adm_grid, cmap='YlGnBu', linewidths=0.3, cbar_kws={'label': 'Admissions'})
plt.title('Hospital Admissions')
plt.xlabel('hours of the day')
plt.ylabel('days of the week')
plt.show()
```



Conclusion



Workforce. The environment is nurse-led, with RNs being the primary caregivers.



Efficiency: Operational response to discharge requests is rapid (typically <1 hour)



Finance: The ICU cost center drives the majority of procedural billing



Clinical: Infectious conditions (Sepsis, Pneumonia) are the dominant reasons for intake



Application: These insights can be used to train Machine Learning models for predicting staffing needs and bed availability